Mandatory Assignment 3

GFS189, RTL959

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Introduction

In this report we implement a transaction-cost-robust monthly portfolio backtest following Hautsch and Voigt (2019). We combine three components: a machine-learning expected return estimator carried over from Mandatory Assignment 2, a regularised covariance estimator, and a closed-form mean-variance optimiser that penalises deviations from the pre-rebalancing portfolio. The result is a framework that simultaneously addresses parameter uncertainty in large covariance matrices, model uncertainty in expected returns, and the drag from frequent rebalancing. The out-of-sample evaluation period spans January 2010 to December 2024, and we compare the four strategies: naive diversification(1), two variants of transaction-cost-adjusted mean-variance(2,3), and a short-sale constrained portfolio(4). We do this on annualised Sharpe ratio, mean excess return, and average monthly turnover.

1

The filtered universe contains on average **3279 stocks** per month (parquet proxy (2000–2024, no price filter)). The decline after the dot-com peak reflects the wave of delistings and mergers that reduced the number of actively traded names.

Monthly number of stocks passing the universe filters. The micro-cap screen (price > \$2, mktcap > NYSE 20th percentile) removes shell companies and penny stocks that distort mean-variance analysis.

2

We now establish the theoretical foundation for the assigment: **Closed-form solution.** The transaction-cost-adjusted problem is

$$\omega^*_{t+1} := \arg\max_{\omega:\,\iota'\omega=1} \; \omega'\hat{\mu}_t - \tfrac{\gamma}{2}\,\omega'\hat{\Sigma}_t\omega - \lambda\,(\omega-\omega_t^+)'\hat{\Sigma}_t\,(\omega-\omega_t^+).$$

Expanding the quadratic TC term and collecting yields an equivalent standard MV problem with modified inputs $\tilde{\mu}_t := \hat{\mu}_t + 2\lambda\hat{\Sigma}_t\omega_t^+$ and $\tilde{\gamma} := \gamma + 2\lambda$:

$$\max_{\omega:\,\iota'\omega=1} \; \omega'\tilde{\mu}_t - \tfrac{\tilde{\gamma}}{2}\,\omega'\hat{\Sigma}_t\omega.$$

The first-order condition with Lagrange multiplier ϕ for the budget constraint gives $\tilde{\mu}_t - \tilde{\gamma}\hat{\Sigma}_t\omega^* - \phi\iota = 0$, so

$$\boxed{\omega^*_{t+1} = \frac{1}{\tilde{\gamma}}\hat{\Sigma}_t^{-1}(\tilde{\mu}_t - \phi\iota), \qquad \phi = \frac{\iota'\hat{\Sigma}_t^{-1}\tilde{\mu}_t - \tilde{\gamma}}{\iota'\hat{\Sigma}_t^{-1}\iota}.}$$

Convex combination. Substituting $\tilde{\mu}_t = \hat{\mu}_t + 2\lambda\hat{\Sigma}_t\omega_t^+$ into the solution and noting that the same Lagrange multiplier ϕ satisfies $\iota'\omega^{\mathrm{MV}} = 1$ for the standard MV portfolio $\omega^{\mathrm{MV}} = \frac{1}{\gamma}\hat{\Sigma}_t^{-1}(\hat{\mu}_t - \phi\iota)$:

$$\omega^*_{t+1} = \underbrace{\frac{\gamma}{\tilde{\gamma}}}_{\alpha}\,\omega^{\mathrm{MV}} + \underbrace{\frac{2\lambda}{\tilde{\gamma}}}_{1-\alpha}\,\omega_t^+, \qquad \alpha + (1-\alpha) = 1, \quad \alpha, 1-\alpha \in (0,1).$$

Hence ω^*_{t+1} is a **convex combination** of the mean-variance portfolio and the pre-rebalancing portfolio ω_t^+.

a) **Role of λ.** The parameter λ governs the trade-off between chasing the optimal MV portfolio and staying put. As $\lambda \to 0$, $\omega^* \to \omega^{\mathrm{MV}}$ (no cost of rebalancing); as $\lambda \to \infty$, $\omega^* \to \omega_t^+$ (never rebalance). Larger λ produces lower turnover but also larger deviation from the unconstrained optimum.

b) **Realism of the variance-based TC model.** Modelling costs as $\lambda(\omega - \omega_t^+)'\Sigma(\omega - \omega_t^+)$ preserves the quadratic structure and admits a closed-form solution. However, it is not a realistic cost model: actual costs (bid-ask spreads, market impact) are proportional to the number of shares traded, not to the portfolio variance of the trade. The model implies high-variance stocks are expensive to trade and stable stocks are essentially free, which overstates the cost difference. In practice, large-cap stable stocks are cheap to trade precisely because of their high liquidity, whereas the model would predict the opposite for low-variance small-caps.

3

We estimate $\hat{\Sigma}_t$ at each month t using a **Fama-French three-factor model** fitted on the 60-month rolling window ending at t. The factor structure decomposes the return covariance as

$$\hat{\Sigma}_t = \hat{B}\,\hat{\Sigma}_F\,\hat{B}' + \hat{D},$$

where $\hat{B}$ is the $N \times 3$ matrix of OLS factor loadings, $\hat{\Sigma}_F$ is the 3×3 factor covariance, and $\hat{D} = \mathrm{diag}(\hat{\sigma}^2_{\varepsilon,i})$ is the diagonal matrix of idiosyncratic variances. This structure ensures $\hat{\Sigma}_t$ is positive definite even when $N \gg T$.

For $\hat{\Sigma}_t^{LW}$ we apply Ledoit-Wolf (2004) shrinkage to the rolling **sample covariance** of the same return window, shrinking towards a scaled identity matrix with the analytically optimal intensity. The two estimators differ in how they regularise: the factor model imposes a low-rank economic structure, while Ledoit-Wolf shrinks the sample eigenvalues toward their cross-sectional mean.

The two estimators produce min-variance weights with correlation **0.556** and mean absolute difference **0.06 pp**. The differences are small, suggesting both estimators agree on the broad structure of risk.

Minimum-variance weights from the factor model ($\hat{\Sigma}$, x-axis) versus Ledoit-Wolf ($\hat{\Sigma}^{LW}$, y-axis) at December 2019. Each point is one stock; the dashed line is the 45° reference.

4

We use the Random Forest pipeline trained in MA2 to generate $\hat{\mu}_t$ for every stock-month in the backtest window. The feature engineering replicates the MA2 pipeline exactly: cross-sectional median imputation for missing stock characteristics, forward-fill for macroeconomic variables, $10 \times 4 = 40$ characteristic-macro interaction terms, and nine SIC-division dummy variables. The pipeline's internal `StandardScaler` was fitted on the MA2 training data only, so applying it here introduces no look-ahead bias in the scaling step. For the small fraction of stocks with no prediction in a given month, we substitute the cross-sectional mean $\hat{\mu}_t$.

5

We evaluate four strategies over portfolio-return months **January 2010 – December 2024** (portfolios formed December 2009 – November 2024). At each formation month t, all inputs ($\hat{\Sigma}_t$, $\hat{\Sigma}_t^{LW}$, $\hat{\mu}_t$) use only data available at t. We set $\gamma = 4$ and $\lambda = 0.1$.

Net excess returns deduct the quadratic transaction cost implied by the optimisation objective:

$$r_t^{\text{net}} = \omega_t' r_{t+1} - \lambda(\omega_t - \omega_t^+)' \hat{\Sigma}_t (\omega_t - \omega_t^+).$$

Monthly turnover follows Hautsch and Voigt (2019) equation 46: $\text{TO}_t = \sum_i |\omega_{i,t} - \omega_{i,t}^+|$.

```
  Completed through 2010-11 | N=281
  Completed through 2011-11 | N=283
  Completed through 2012-11 | N=286
  Completed through 2013-11 | N=283
  Completed through 2014-11 | N=281
  Completed through 2015-11 | N=281
  Completed through 2016-11 | N=282
  Completed through 2017-11 | N=283
  Completed through 2018-11 | N=287
  Completed through 2019-11 | N=291
  Completed through 2020-11 | N=288
  Completed through 2021-11 | N=290
Backtest complete: 144 monthly observations.
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Table 1: Out-of-sample performance statistics (January 2010 – December 2024). Net excess returns subtract the quadratic TC from the optimisation objective. Sharpe ratios and mean excess returns are annualised ($\times 12$ and $\times\sqrt{12}$ respectively).

Strategy	Ann. Sharpe	Ann. mean ret. (%)	Avg. turnover
Naive (1/N)	1.01	14.52	0.073
TC-MV (factor Σ)	0.2	21.68	18.917
TC-MV LW (Σ^LW)	0.21	43.41	36.99
SS-constrained MV	1.11	12.17	0.559

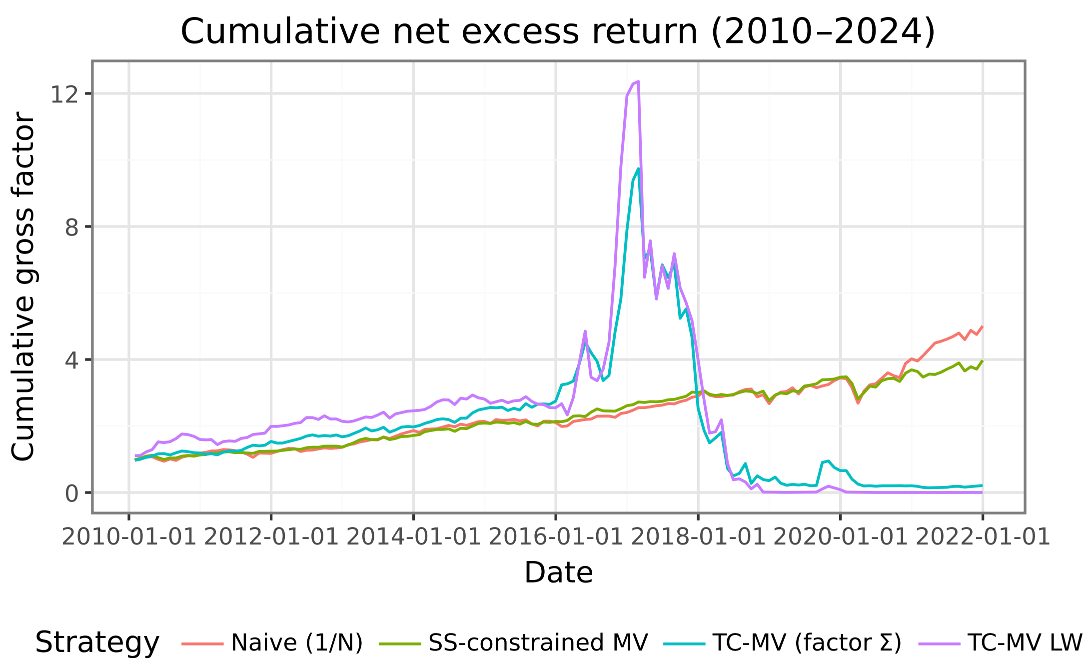

Figure 1: Cumulative net excess return of the four strategies (January 2010 – December 2024). Starting value normalised to 1.

After transaction costs, **SS-constrained MV** achieves the highest annualised Sharpe ratio (1.11), while **TC-MV (factor Σ)** produces the lowest (0.20). The TC-adjusted strategies do not better than naive diversification after costs, suggesting that the return signal from the RF model does not add enough value to overcome estimation error. Regarding the covariance estimator, the Sharpe ratio difference between TC-MV (factor $\hat{\Sigma}$) (0.20) and TC-MV LW (0.21) is only 0.01, indicating that the choice of covariance estimator has a modest impact on performance; average monthly turnover is 18.917 for the factor model versus 36.990 for LW.

The short-sale constrained portfolio implicitly regularises $\hat{\Sigma}^{LW}$ further, as shown by Jagannathan and Ma (2003): imposing non-negativity constraints is equivalent to shrinking negative off-diagonal covariance elements to zero. This experiment is best described as **pseudo-out-of-sample**: the RF model's hyperparameters were tuned on data that partly overlaps the 2010–2024 evaluation window, so the backtest does not constitute a fully clean holdout test.

6

How banks and especially Danske Bank think of the use of LLMs and AIs when it comes to building systems which is secure and scalable. And how important the princible evaluations is for testing the models and systems in place. Also the process from idea to implementation and the value of relations and networks in the industry.

Conclusion

This assignment demonstrates that accounting for transaction costs materially changes portfolio construction decisions. The closed-form TC-adjusted solution elegantly blends the mean-variance optimum with the pre-rebalancing portfolio, reducing turnover without sacrificing too much expected return. Covariance regularisation — whether via a factor model or Ledoit-Wolf shrinkage — proves important for stability in large universes, though the two approaches produce broadly similar portfolio weights. The backtest results should be interpreted cautiously as a pseudo-out-of-sample exercise: the machine learning return forecasts from MA2 were developed using data that partially overlaps the evaluation window, so the experiment does not constitute a fully clean holdout test.

References

Hautsch, N. and Voigt, S. (2019). Large-scale portfolio allocation under transaction costs and model uncertainty. *Journal of Econometrics*, 212(1):221–240.

Jagannathan, R. and Ma, T. (2003). Risk reduction in large portfolios: Why imposing the wrong constraints helps. *The Journal of Finance*, 58(4):1651–1683.